KEYNOTE (GREEN)

药物发现中的数据驱动毒性预测：现状与未来方向

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早期毒性评估在药物发现过程中起着至关重要的作用，因为它对候选药物的淘汰率有重要影响。最近，信息技术的不断升级极大地推动了毒性预测的持续发展。为了概览数据驱动毒性预测的现状，我们回顾了相关研究，并从三个主要方面进行了总结：毒性预测的特征和难点、建模方法的演变以及可用的毒性预测工具。在每个部分，我们阐述了研究现状、存在的挑战及可行的解决方案。最后，还提出了毒性预测的若干新方向和建议。

关键词: 毒性预测; 机器学习; 深度学习; 预测工具

介绍

药物研究与开发（R&D）是一个“知识密集型”的过程，其特点是高风险、高投入和周期长。尽管在制药技术方面已取得了很大进展，但药物候选者的淘汰率仍然很高，估计为96%，其中30%的淘汰是由于缺乏安全性。[p1],[p2],[p3],[p4] 有无数潜在的临床试验实体，即使在上市后，有些也因毒性不可接受而不得不撤回。环境科学是另一个受化学合物毒性影响的领域，因为任何物质在投入使用之前都必须满足监管机构的毒性测试标准，以确保其不会对环境和生物造成危害。因此，需要对新化学实体进行广泛的安全评估。

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KEYNOTE (GREEN)

Data-driven toxicity prediction in drug discovery: Current status and future directions

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Early toxicity assessment plays a vital role in the drug discovery process on account of its significant influence on the attrition rate of candidates. Recently, constant upgrading of information technology has greatly promoted the continuous development of toxicity prediction. To give an overview of the current state of data-driven toxicity prediction, we reviewed relevant studies and summarized them in three main respects: the features and difficulties of toxicity prediction, the evolution of modeling approaches, and the available tools for toxicity prediction. For each part, we expound the research status, existing challenges, and feasible solutions. Finally, several new directions and suggestions for toxicity prediction are also put forward.

Keywords: toxicity prediction; machine learning; deep learning; predictive tools

Introduction

Drug research and development (R&D) is a 'knowledge-concentrated' process characterized by high risk, high investment, and long periods. Although much progress has been made in pharmaceutical technology, it is estimated that the attrition rate of drug candidates remains high, at 96%, with 30% of the attrition being due to a lack of safety.[p1],[p2],[p3],[p4] There are countless potential entities in clinical trials, and even after marketing some have had to be withdrawn due to unacceptable toxicity. Environmental science is another area affected by the toxicity of chemical compounds, as any substance must meet the toxicity test standards of regulatory agencies before it is put into use to ensure that it does not cause harm to the environment and organisms. Therefore extensive safety evaluation of new chemical entities

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应尽早在药物开发和环境安全评估过程中进行。

早期的安全性评估主要依赖传统的体内和体外方法，由于其固有缺陷，如高成本、伦理限制以及耗时耗力等，在应用上存在局限性。随后，随着计算技术和跨学科方法的普及，体外计算方法（in silico methods）如结构警示（structural alerts）、类比推断（read-across）以及定量构效关系（QSARs）分析在药物设计早期的毒性评估中显示出了独特优势，包括绿色、快速、廉价和高准确性；最重要的是，它们可以在化合物合成之前应用。关于QSAR分析，自1962年这一科学概念正式提出以来，各种建模方法不断发展，以处理日益增长的毒性数据。其中，机器学习（ML）已成为发展最快且使用最广的方法，作为

(p13),(p14),(p15) 通常，使用机器学习 (ML) 的数据驱动毒性预测研究旨在构建分类或回归模型，以根据从实验数据中获得的知识描述分子化学结构与其毒性终点之间的复杂关系。[(p16),(p17),(p18)] 近几年的连续毒性预测研究和多篇综述已经证实了机器学习方法在毒性预测领域的实用性。[(p5),(p8),(p14),(p15)] 此外，最近由于硬件的改进和极大规模数据集的可用性，机器学习的一个子领域——深度学习 (DL) 已在毒性评估领域超越了传统的机器学习方法。[(p19),(p20),(p21)]

近年来，信息技术的不断升级推动了毒性预测的持续发展。为了全面了解药物发现和环境科学中基于数据的毒性预测，本文回顾了现有的基于机器学习的研究，并从以下三个方面进行总结（图1）：(1) 数据驱动毒性预测的特征和困难，包括数据集小、机制复杂以及数据质量低等问题。(2) 建模方法的演变，从传统机器学习到深度学习及后深度学习时代，主要关注各种学习方法的优缺点、深层关系及方法之间的差异。(3) 毒性预测的流行软件和工具，主要讨论所覆盖的终点、建模方法、预测准确性及解释能力。对于(1)–(3)，本文还讨论了现存问题及可行的解决建议。

基于数据的毒性预测的特征与难点

在过去几十年里，计算机辅助药物设计（CADD）经历了快速发展，而作为其中重要组成部分的毒性评估也引起了广泛关注。随着计算能力的不断提升和新理论的提出，越来越多的CADD研究人员尝试提出有效方法来构建

针对各种毒性终点，我们构建了回归或分类模型，并取得了一些令人满意的结果。然而，对于某些毒性终点，即使使用了各种机器学习方法，预测性能仍然普遍处于较低水平。为了找出潜在原因并全面理解数据驱动毒性预测的特征和难点，我们整理了先前曾被预测过的45个毒性终点，并在图2中列出了它们的信息。基于该图，我们总结了毒性预测任务的几个特征如下。

相对较小的数据集

为了探索毒性数据集的潜在特征，我们收集了覆盖45个不同终点的50个数据集，并绘制了饼图来表示它们（图2a）。从该图可以看出，只有34%的数据集（17/50）包含超过5000个化合物，且只有一个数据集（hERG）包含超过10,000个化合物。具体来说，大多数包含足够化合物的数据集（12/17）是通路毒性数据集、人体以太相关基因（Tox21）数据集，其余的数据集通常对应由hERG阻断剂、Ames诱变毒性、细胞毒性和大鼠口服急性毒性代表的一类毒性终点。由于其独特的毒理学意义、明确的终点、相对简单的机制以及充足的实验数据，这些数据集在早期被选中用于研究其应用。

复杂的机制

从现代医学的角度来看，大多数毒性终点具有复杂的作用机制，涉及生理学、病理学和药理学。机制的复杂性体现在两个方面：未知机制和多重机制。具有未知机制的毒性终点通常过于异质和复杂，缺乏明确的解释。

should be conducted as early as possible in the drug development and environmental safety evaluation process.[(p5)]

Early safety assessment mainly relies on traditional *in vivo* and *in vitro* methods, which are limited in terms of application due to inherent shortcomings such as expense, ethical restrictions, and their time- and labor-consuming nature.[(p6),(p7),(p8)] Subsequently, with the popularity of computational technology and interdisciplinary methods, *in silico* methods such as structural alerts, read-across, and analysis of quantitative structure–activity relationships (QSARs) have shown unique advantages for toxicity assessment in the early stages of drug design, including their green, fast, cheap, and accurate nature; most importantly, they can be applied before a compound is synthesized.[(p9),(p10)] Regarding QSAR analysis, since this scientific concept was formally proposed in 1962,[(p11),(p12)] various modeling methods have been developed to process the increasing amount of toxicity data. Among them, machine learning (ML) has become the fastest-growing and most widely used method, serving as an important component of data-driven toxicity prediction research.[(p5),(p8),(p13),(p14),(p15)] Generally, data-driven toxicity prediction studies using ML aim to build classification or regression models to describe the complex relationships between the chemical structure of molecules and their toxicity endpoints based on knowledge obtained from experimental data.[(p16),(p17),(p18)] Continuous toxicity prediction studies and several reviews in the past few years have confirmed the utility of ML methods in the field of toxicity prediction.[(p5),(p8),(p14),(p15)] Furthermore, recently, owing to hardware improvements and the availability of extremely large datasets, deep learning (DL), a subfield of ML, has surpassed traditional ML methods in the toxicity assessment field.[(p19),(p20),(p21)]

In recent years, constant upgrading of information technology has promoted the continuous development of toxicity prediction.[(p8),(p22),(p23)] To get a full picture of data-driven toxicity prediction in drug discovery and environmental science, we review the available ML-based studies in this paper and summarize them in the following three respects Figure 1. (1) The features and difficulties of data-driven toxicity prediction, including small datasets, complex mechanisms, and low-quality data. (2) The evolution of modeling methods, from traditional ML to DL and the post-DL era, mainly focusing on the advantages and disadvantages of various learning methods, deep relationships, and the differences between methods. (3) Popular software and tools for toxicity prediction, mainly discussing the endpoints covered, modeling methods, prediction accuracy, and explanatory ability. For (1)–(3), the existing issues and feasible suggestions to overcome them are discussed.

The features and difficulties of data-driven toxicity prediction

In the past few decades, computer-aided drug design (CADD) has undergone rapid development, and toxicity evaluation, as an important part of this, has also attracted extensive attention. With the continuous improvement of computing power and the proposal of new theories, more and more CADD researchers have attempted to derive effective methods to construct regression or classification models for a variety of toxicity endpoints and have achieved some satisfactory results. However, for some toxicity endpoints, prediction performance is consistently at a lower level even though various ML methods have been used. To identify the underlying reasons and comprehensively understand the features and difficulties of data-driven toxicity prediction, we collated 45 toxicity endpoints that had previously been predicted and list their information in Figure 2. Based on this figure, we summarize several features for toxicity prediction tasks as follows.

Relatively small datasets

To explore the underlying features of toxicity datasets, we collected 50 datasets covering 45 different endpoints and drew a pie diagram to represent them (Figure 2a). From this figure, we can see that only 34% of datasets (17/50) contain more than 5000 compounds, and only one dataset (hERG) has more than 10,000 compounds. Specifically, most of the datasets (12/17) containing sufficient compounds (>5000) are pathway toxicity datasets [human ether-a-go-go related gene (Tox21) datasets],[(p24),(p25),(p26),(p27)] and the remaining datasets generally correspond to a class of toxicity endpoints represented by hERG blockers,[(p28),(p29),(p30),(p31)] Ames mutagenicity toxicity,[(p32),(p33),(p34),(p35),(p36),(p37)] cytotoxicity,[(p38),(p39),(p40),(p41)] and oral rat acute toxicity.[(p42),(p43),(p44),(p45)] Owing to their unique toxicological significance, definite endpoints, relatively simple mechanism, and sufficient experimental data, the datasets were selected for studying, at an early stage, the application of ML to toxicity prediction, and subsequent intensive research made them well-known in this field. However, most toxicity datasets are relatively small (<5000, 33/50), with a large proportion having fewer than 500 compounds (11 of 50, 22%). In recent years, with the release of more and more toxicity databases, the data on some toxicity endpoints have been slightly supplemented (e.g. developmental toxicity,[(p46),(p47),(p48)] hepatotoxicity,[(p49),(p50),(p51)] and genotoxicity[(p52),(p53),(p54)]), but there are still some complex toxicity endpoints lacking data, such as nephrotoxicity,[(p55),(p56)] rhabdomyolysis,[(p57)] hematotoxicity,[(p58),(p59)] and so on.[(p60),(p61),(p62),(p63),(p64),(p65)] From the perspective of mathematical modeling, a smaller dataset carries less information, resulting in poor predictive ability of the output model; this cannot be significantly improved even by using advanced modeling methods, and several recent studies using different modeling methods have shown that DL does not perform as well as expected in predicting some specific toxicity endpoints.[(p50),(p58),(p65)] In summary, a relatively small dataset greatly limits the prediction of toxicity endpoints, which is a major obstacle to the development of current and future data-driven toxicity assessments.

Complex mechanism

From the point of view of modern medicine, most toxicity endpoints have complex mechanisms of action, involving physiology, pathology, and pharmacology. The complexity of the mechanisms is reflected in two aspects: unknown mechanisms and multiple mechanisms. Toxicity endpoints with unknown mechanisms are usually too heterogeneous and complex, lacking

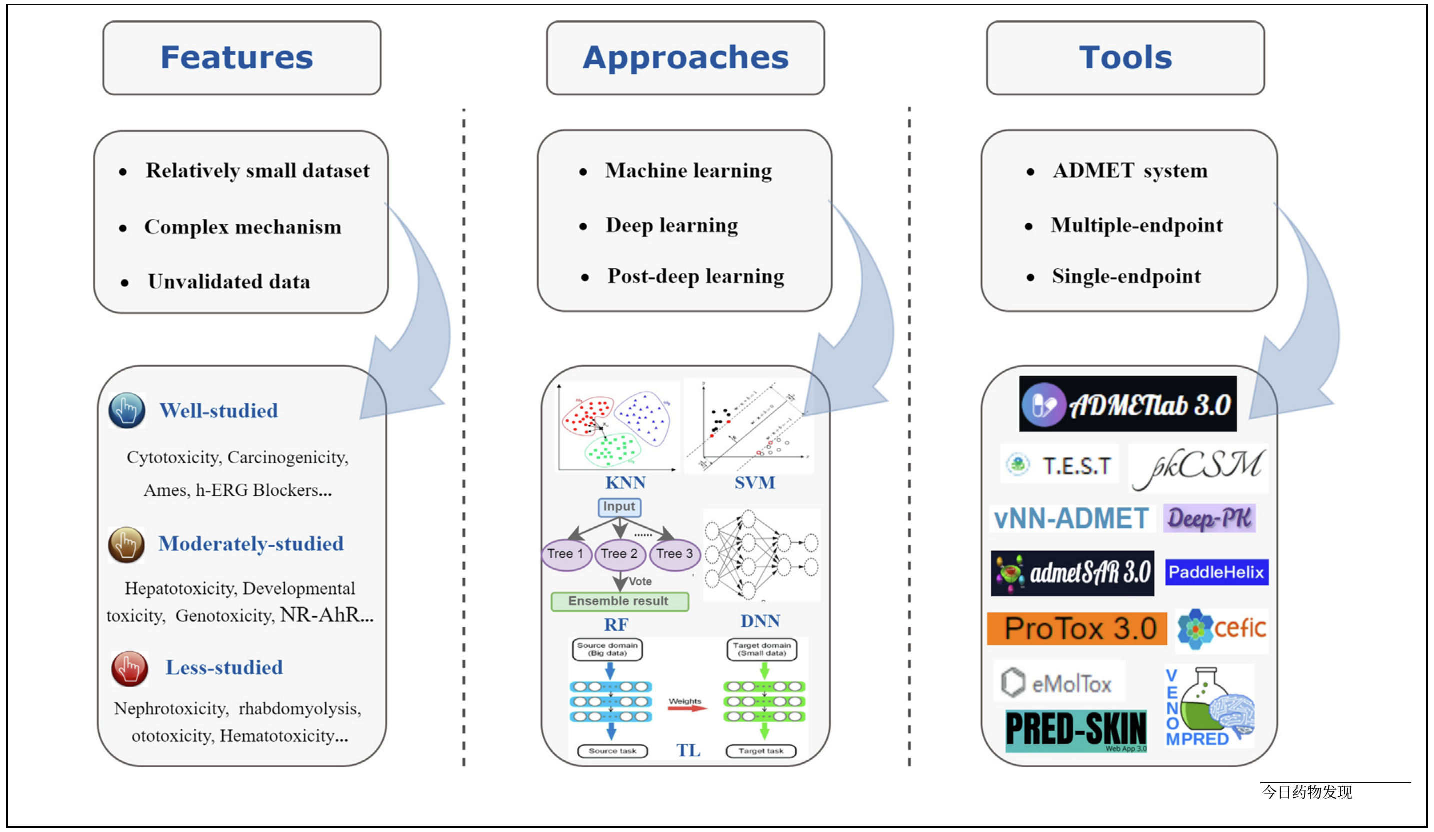

图 1
数据驱动的毒性预测概述。该示意图由三部分组成。第一部分涉及数据驱动毒性预测的特征和困难，包括数据集相对较小、毒性机制复杂以及数据未经验证。第二部分涉及毒性预测方法，包括机器学习、深度学习及深度学习后的方法。最后，列出了当前可用的毒性预测工具，包括ADMET系统和针对特定毒性预测的工具。

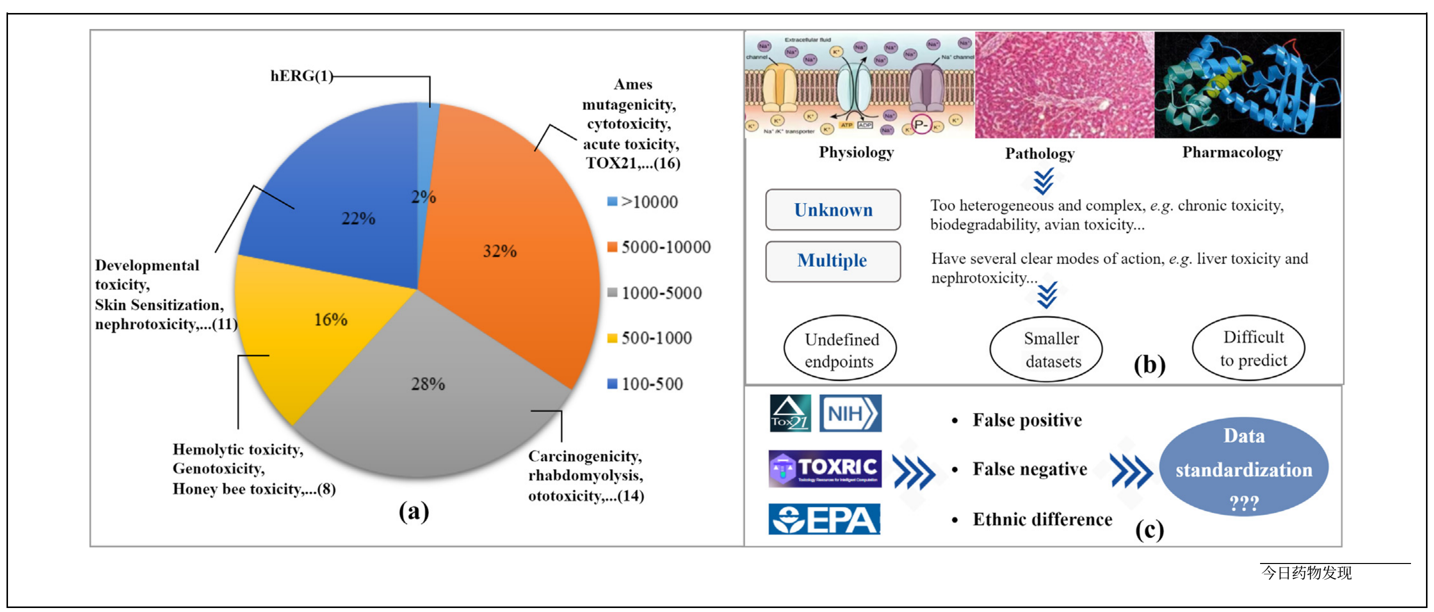

图2
毒性预测的特点 (a) 数据集相对较小。饼图显示大多数毒性数据集相对较小（<5000，33/50），其中一些数据集包含的化合物少于500种（50个中的11个，占22%）。(b) 化学物质毒性的复杂机制。大多数毒性终点具有复杂的作用机制，涉及生理学、病理学和药理学，这体现在两个方面：机制未知和多重机制；(c) 现有毒性数据的问题，包括假阳性数据、假阴性数据，以及动物实验和人类数据之间的差异，导致毒性预测存在偏差。

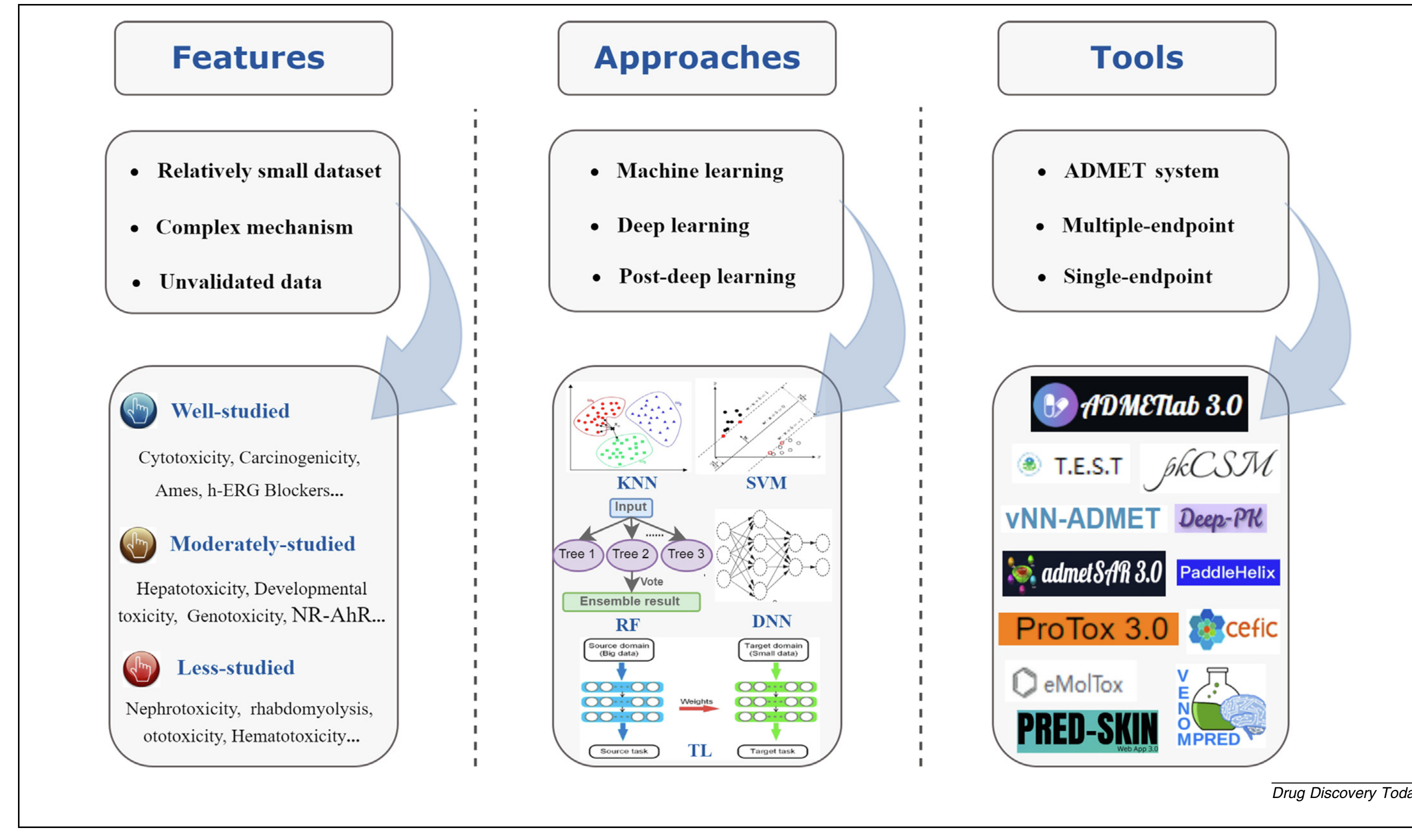

FIGURE 1
Overview of the data-driven toxicity prediction. This schematic consists of three parts. The first part pertains to the features and difficulties of data-driven toxicity prediction, including relatively small datasets, complex toxicity mechanisms, and unvalidated data. The second part pertains to toxicity prediction methods, including machine learning, deep learning and post-deep learning methods. Finally, currently available toxicity prediction tools are listed, including the ADMET system and prediction tools for specific toxicities.

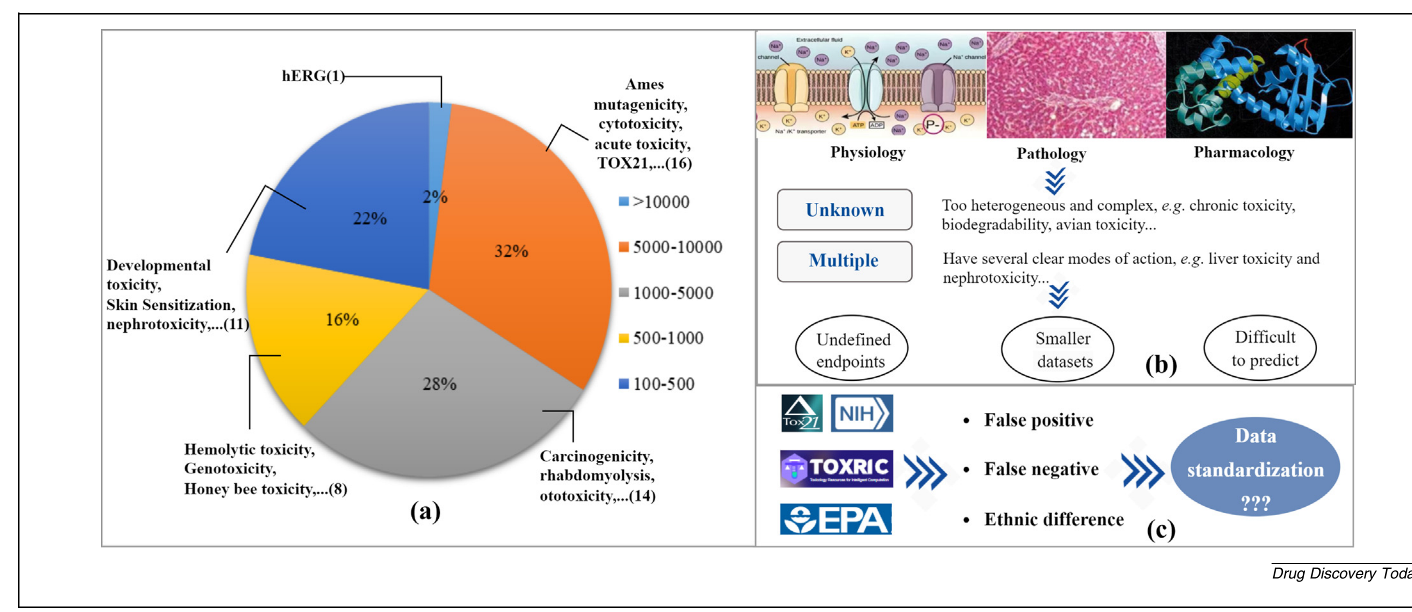

FIGURE 2
Features of toxicity prediction **(a)** Relatively small datasets. The pie diagram shows that most of the toxicity datasets are relatively small (<5000, 33/50), and some contain fewer than 500 compounds (11 of 50, 22%). **(b)** The complex mechanism of chemical toxicity. Most toxicity endpoints have complex mechanisms of action encompassing physiology, pathology, and pharmacology, which is reflected in two aspects: unknown mechanisms and multiple mechanisms; **(c).** Existing problems with toxicity data, including false-positive data, false-negative data, and the differences between animal tests and human data, have led to bias in toxicity predictions.

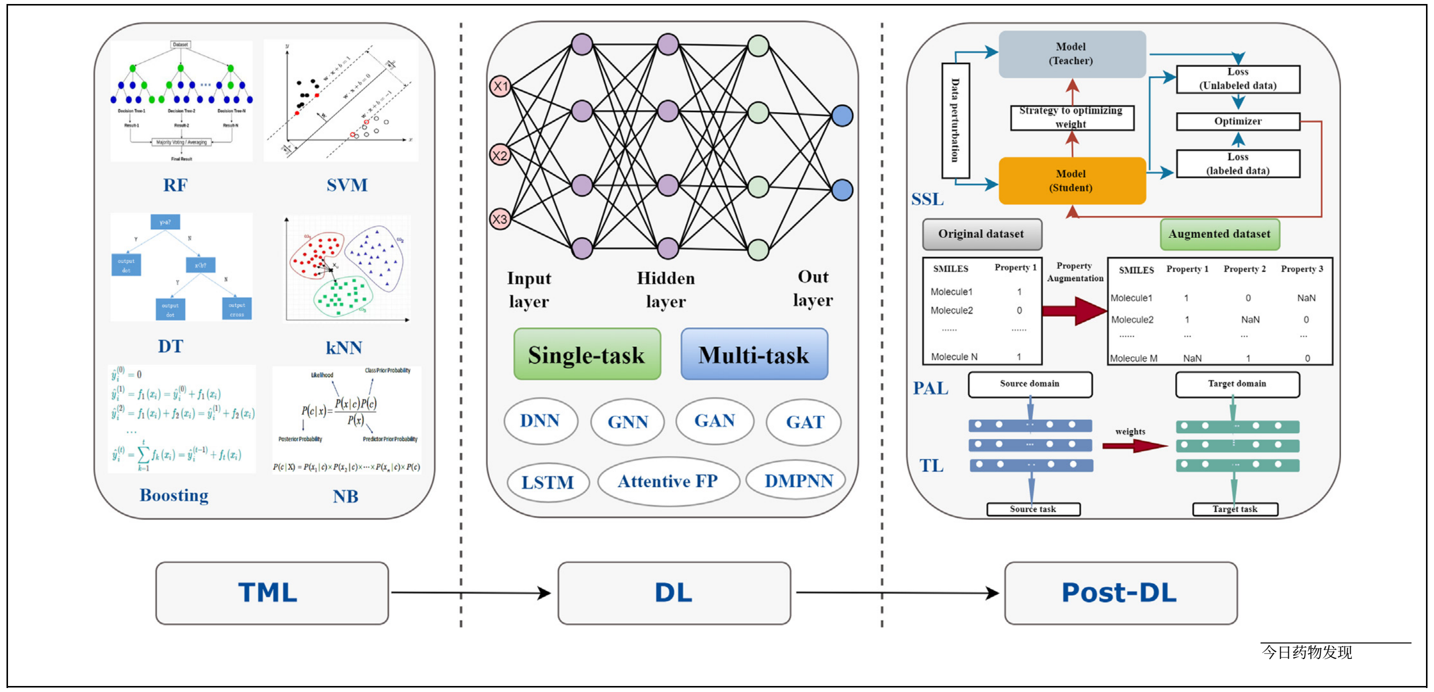

图3
在毒性预测中常用的计算方法主要有三类：传统机器学习，包括随机森林、支持向量机和决策树；深度学习，包括深度神经网络、卷积神经网络和循环神经网络；以及后深度学习时代的方法，包括半监督学习、迁移学习和性质增强学习。

一种易于理解的机制或易于解释的能力，例如在慢性毒性、生物降解性、鸟类毒性等方面的解释。另一个方面是具有多种机制的毒性终点，这些终点具有几个明确的作用模式，以肝毒性和肾毒性为代表。以肝毒性为例，由于其复杂的潜在机制，如直接肝细胞毒性、免疫相关损伤、线粒体损伤和炎症反应，其计算预测仍然极具挑战性。因此，尽管已经进行了大量工作以尝试准确预测药物性肝损伤（DILI），但结果并不十分理想。通常，毒性终点的复杂机制会导致终点未明确以及数据集较小，进而增加预测难度并降低预测能力。因此，数据驱动的预测...

低质量数据

对于基于数据的毒性预测，数据的质量仍然是一个严重且根本性的问题。目前，大多数可用的毒性数据是通过高通量体外检测或动物体内测试收集的，例如 Tox21 和 ToxCast。

尽管它们为基于机器学习的毒性预测提供了进一步发展的机会，但不可否认的是，仍然存在一些问题，包括假阳性数据、假阴性数据以及动物数据与人类数据之间的差异，这些都会对毒性预测引入偏差。这也是大多数QSAR模型预测结果与临床实际情况存在较大差异的主要原因。为改善低质量毒性数据的问题，一个显而易见的方法是收集更多可靠的临床数据，另一个有效的解决方案是对现有的毒性数据进行标准化。尽管已经有许多包含临床毒性数据的数据库，比如SIDER、LiverTox、PharmGKB和Drugs.com数据库，但我们仍需要一个开放访问的基准数据库，该数据库应具备明确且准确的毒性终点描述，甚至包括毒性的不良结果途径（AOP）。来自不同国家实验室的毒性数据的质量和r

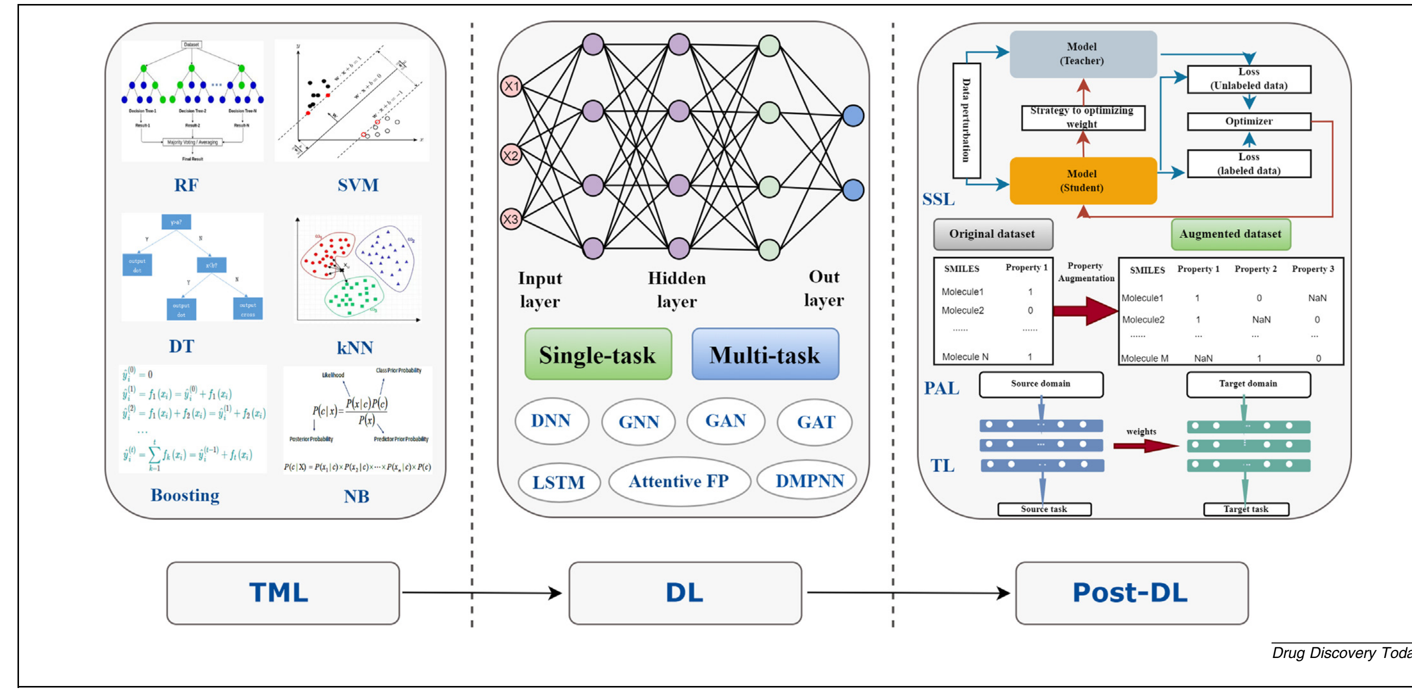

FIGURE 3
Computational methods commonly used in toxicity prediction. There are three main categories: traditional machine learning, including random forest, support vector machine, and decision tree; deep learning, including deep neural networks, convolutional neural networks, and recurrent neural networks; and the post-deep learning era, including semi-supervised learning, transfer learning, and property augmentation learning.

an easily understandable mechanism or to the ability to be well explained, such as in terms of chronic toxicity, biodegradability, avian toxicity, etc.[p25],[p60],[p61],[p66] The other aspect is toxicity endpoints with multiple mechanisms, which have several well-defined modes of action, represented by liver toxicity and nephrotoxicity.[p50],[p51] Taking hepatotoxicity as an example, its computational prediction is still extremely challenging due to the complex underlying mechanisms, such as direct hepatocytotoxicity, immune-related damage, mitochondrial damage, and the inflammatory response.[p49],[p68] Therefore although a great deal of work has been done to try to accurately predict drug-induced liver injury (DILI), the results are not very satisfactory. Generally, the complex mechanism of toxicity endpoints will lead to undefined endpoints and smaller datasets, which will further result in increased prediction difficulty and decreased predictive ability. Therefore the data-driven prediction of some toxicities is a difficult task due to their inherently complex mechanisms. Based on the current research, we hold the opinion that collecting more well-validated data and developing more elaborate prediction models with distinct toxicity mechanisms or endpoints may be a feasible direction for the future prediction of toxicity endpoints with complex mechanisms.

Low-quality data

For data-driven toxicity prediction, the quality of the data is still a serious and fundamental issue. At present, most available toxicity data are collected through high-throughput *in vitro* assays or *in vivo* tests on animals, such as Tox21 and ToxCast.[p69] Although they provide an opportunity for further development of toxicity prediction based on ML, it is undeniable that there are still some problems including false-positive data, false-negative data, and the differences between animal and human data, which introduce bias into toxicity prediction. This is the main reason why there is a large difference between the results predicted by most QSAR models and the clinical reality. To improve the problem of low quality toxicity data, one obvious method is to collect more reliable clinical data, and another effective solution is to standardize the available toxicity data. Although there are already many databases containing clinical toxicity data, such as the SIDER, LiverTox, PharmGKB, and Drugs.com databases, we need an open-access benchmark database with clear and accurate descriptions of toxicity endpoints and even the adverse outcome pathway (AOP) of toxicity. The quality of toxicity data from laboratories in different countries and regions is highly variable, and the low interlaboratory repeatability of the data will lead to a phenomenon whereby, as modeling methods improve, the prediction accuracy of a given model will reach saturation such that, even if more test data can be accumulated, further improvement in predictive power is difficult.[p70],[p71],[p72],[p73],[p74],[p75] This is also a major problem in terms of current data collection, which fundamentally hinders the continuous optimization of toxicity prediction models. Therefore from the point of view of toxicity data, there are two main future directions: (1) collect and screen more high-quality clinical toxicity data to improve the practicability of the data; and (2) develop a standardized process for toxicity

数据验证并建立一个大型基准数据库，仅包含经过充分验证的测试结果，以减少实验室间偏差。

毒性预测的建模方法

在数据驱动的毒性预测过程中，建模方法的选择是一个核心且关键的问题。在过去的几十年里，计算机技术的发展促进了各种建模方法的出现，本综述将其大致分为三类/三个阶段：传统机器学习（TML）、深度学习（DL）以及深度学习后时代（PDL）。不同的建模方法源于不同的实际需求，并具有不同的特征和应用领域，如图3所示。

TML

在毒性预测的早期阶段，首先采用了统计分析方法，如多元线性回归（MLR）和偏最小二乘法（PLS），以探讨化学结构与毒性之间可能的关系，但生物系统的复杂性以及实验数据量的增加使得这些方法在许多情况下不尽如人意。此时，TML 作为一类能够建模更复杂任务的先进算法，吸引了毒理学家的关注。毒性预测研究中常用的 TML 方法主要包括支持向量机（SVM）、随机森林（RF）、决策树（DT）、k 近邻（kNN）、朴素贝叶斯（NB）等。它们的基本概念、统计特性、优点和缺点已在若干前期综述中得到充分讨论。[(p5),(p8),(p14),(p15)]

近年来，随着 TML 方法的日益成熟以及毒性数据的不断增加，研究人员比较了这些流行的 TML 方法，以找出最适合每种特定毒性的方法。[(p46),(p52),(p55),(p56),(p57),(p61),(p62),(p63),(p66),(p67),(p76)] 我们在表1中简要总结了它们的详细信息，包括12个毒性终点、所使用的TML方法、最佳方法、最终模型的预测能力、应用领域以及原始参考文献。在这些方法中，SVM在大多数毒性终点上表现最佳，其次是RF。这似乎是直观的。SVM，在特定任务背景下也被称为支持向量分类器（SVC）或支持向量回归（SVR），以其高预测性能和低过拟合风险而闻名。在毒性预测的背景下，SVM算法使用核函数，如线性、多项式和Sigmoid函数，以及径向基函数（RBF），将由描述符向量编码的分子投射到一个空间中，以最大化不同类别之间的间隔（或分离/边界）。考虑到与小数据集、复杂机制和混合信息相关的毒性预测困难，

表格 1

不同 TML 方法在毒性预测中的比较

毒性终点	研究了TML方法	最佳 TML 方法	内部验证	外部验证	应用领域	Refs
基因毒性	SVM、朴素贝叶斯、k近邻、决策树、随机森林、人工神经网络	SVM	ACC = 0.889	ACC = 0.908	药物发现	(第52页)
尿路毒性	RVM、SVM、随机森林、决策树、XGBoost、AdaBoost、SVMBoost，RVMBoost	SVMBoost	AUC = 1.000 q^2_{LOO} = 0.904	AUC = 0.893; q^2_{ext} = 0.845	药物发现	(第56页)
横纹肌溶解症	kNN，人工神经网络（ANN），支持向量机（SVM），快速分阶段多变量线性回归，多个线性回归，偏最小二乘回归，随机森林，深度神经网络	RF	ACC = 0.793	ACC = 0.753	药物发现	(第57页)
鸟类毒性	支持向量机 (SVM)、决策树 (DT)、k近邻 (kNN)、随机森林 (RF)、朴素贝叶斯 (NB)	SVM	ACC = 0.866	ACC = 0.851	环境的安全	(第61页)
溶血性毒性	k近邻 (kNN), 支持向量机 (SVM), 随机森林 (RF), 梯度提升机 (GBM)	SVM	Q^2 = 0.653	–	药物发现	(第62页)
线粒体毒性	kNN，LR，RF，SVM，XGBoost	RF	AUC = 0.899	AUC = 0.921	药物发现	(第63页)
蜜蜂毒性	人工神经网络 (ANN)、决策树 (DT)、k近邻 (kNN)、朴素贝叶斯 (NB)、随机森林 (RF)、支持向量机 (SVM)	SVM	ACC = 0.890	ACC = 0.904	环境的安全	(第64页)
可生物降解性	支持向量机 (SVM)、决策树 (DT)、k最近邻 (kNN)、朴素贝叶斯 (NB)	SVM	AUC = 0.891	AUC = 0.844	环境的安全	(第66页)
肾毒性	人工神经网络, 支持向量机	SVM	ACC = 0.969	ACC = 0.933	药物发现	(第67页)
水生毒性	k近邻 (kNN), 支持向量机 (SVM), 人工神经网络 (ANN), 随机森林 (RF), 朴素贝叶斯 (NB), 决策树 (DT)	SVM	ACC = 0.760	ACC = 0.850	环境的安全	(第76页)
神经毒性	BGR、ETR、GPR、kNN、MLPR、Nu-SVR、RF、SVR	ETR	q^2_{test} = 0.828	q^2_{test} = 0.784	药物发现	(第55页)
发育毒性	朴素贝叶斯, 支持向量机, 随机投影, k近邻, 决策树, 随机森林, AdaBoost	NB	ACC = 0.935; Q^2 = 0.728	ACC = 0.891; R^2 = 0.697	药物发现	(第46页)

缩写：ACC，准确率；ANN，人工神经网络；XGBoost，极端梯度提升；GBM，梯度提升机；LR，逻辑回归；BGR，袋装回归器；ETR，额外树回归器；GPR，高斯过程回归；RP，递归分区；RVM，相关向量机。

data validation and establish a large benchmark database consisting only of well-validated test results to reduce interlaboratory bias.

Modeling approach for toxicity prediction

In the process of data-driven toxicity prediction, the choice of modeling approach is a central and crucial issue. Over the past few decades, the development of computer technology has facilitated the emergence of various modeling approaches, which are roughly classified into three types/stages in this review: traditional machine learning (TML), DL, and the post-deep learning era (PDL). Different modeling methods are derived from different practical requirements and have different characteristics and application fields Figure 3.

TML

In the early stage of toxicity prediction, statistical analysis methods, such as multiple linear regression (MLR) and partial least squares (PLS), were first employed to explore the possible relationships between chemical structures and toxicities, but the complexities of biological systems and the increasing volume of experimental data make these methods unsatisfactory in many cases. At this time, TML attracted the attention of toxicologists as a class of advanced algorithms that can model more complex tasks. Popular TML methods in toxicity prediction studies mainly include support vector machine (SVM), random forest (RF), decision tree (DT), k-nearest neighbor (kNN), naïve bayes (NB), etc. Their basic concept, statistical characteristics, advantages, and disadvantages have been fully discussed in several previous reviews.[(p5),(p8),(p14),(p15)]

In recent years, with the increasing maturity of TML methods and the continuous increase in toxicity data, researchers have compared these popular TML methods to find the one most suitable for each particular toxicity.[(p46),(p52),(p55),(p56),(p57),(p61),(p62),(p63),(p66),(p67),(p76)] We briefly summarize their details in Table 1, including 12 toxicity endpoints, the TML methods applied, the optimal method, the predictive ability of the final model, the application area, and the original references. Among these methods, SVM performed best for most toxicity endpoints, followed by RF. This seems intuitive. SVM, also known as support vector classifier (SVC) or support vector regression (SVR) in the context of particular tasks, is well known for its high predictive performance and low risk of overfitting. In the context of toxicity prediction, the SVM algorithm uses kernel functions, such as linear, polynomial, and sigmoid functions, as well as the radial basis function (RBF), to project molecules encoded by descriptor vectors into a space that maximizes the margin (or separation/boundary) between different classes.[(p77)] Considering the difficulties of toxicity prediction associated with small datasets, complex mechanisms, and mixed information, SVM, as a supervised TML algorithm to handle datasets with high-dimensional variables, will be the optimal solution in many cases. As for RF, an ensemble modeling method based on building multiple DTs as the base learner is a nonlinear multiclass ML method.[(p15)] Similar to SVM, one of the main advantages of RF is that it can handle a large number of features without overfitting and is insensitive to data with missing values and noise. Other TML methods, including naïve Bayes (NB), kNN, the artificial neural network (ANN), extreme gradient boosting (XGBoost), etc., have also been applied to model toxicity data in these studies, but their performance is not as satisfactory. To sum up, although

TABLE 1

Comparison of different TML methods for toxicity prediction

Toxicity endpoint	Studied TML methods	Optimal TML method	Internal validation	External validation	Application area	Refs
Genotoxicity	SVM, NB, kNN, DT, RF, ANN	SVM	ACC = 0.889	ACC = 0.908	Drug discovery	(p52)
Urinary tract toxicity	RVM, SVM, RF, DT, XGBoost, AdaBoost, SVMBoost, RVMBoost	SVMBoost	AUC = 1.000 q^2_{LOO} = 0.904	AUC = 0.893; q^2_{ext} = 0.845	Drug discovery	(p56)
Rhabdomyolysis	kNN, ANN, SVM, fast stagewise multivariate linear regression, MLR, PLS, RF, deep neural network	RF	ACC = 0.793	ACC = 0.753	Drug discovery	(p57)
Avian toxicity	SVM, DT, kNN, RF, NB	SVM	ACC = 0.866	ACC = 0.851	Environmental safety	(p61)
Hemolytic toxicity	kNN, SVM, RF, GBM	SVM	Q^2 = 0.653	–	Drug discovery	(p62)
Mitochondrial toxicity	kNN, LR, RF, SVM, XGBoost	RF	AUC = 0.899	AUC = 0.921	Drug discovery	(p63)
Honey bee toxicity	ANN, DT, kNN, NB, RF, SVM	SVM	ACC = 0.890	ACC = 0.904	Environmental safety	(p64)
Biodegradability	SVM, DT, kNN, NB	SVM	AUC = 0.891	AUC = 0.844	Environmental safety	(p66)
Nephrotoxicity	ANN, SVM	SVM	ACC = 0.969	ACC = 0.933	Drug discovery	(p67)
Aquatic toxicity	kNN, SVM, ANN, RF, NB, DT	SVM	ACC = 0.760	ACC = 0.850	Environmental safety	(p76)
Neurotoxicity	BGR, ETR, GPR, kNN, MLPR, Nu-SVR, RF, SVR	ETR	q^2_{test} = 0.828	q^2_{test} = 0.784	Drug discovery	(p55)
Developmental toxicity	NB, SVM, RP, kNN, DT, RF, AdaBoost	NB	ACC = 0.935; Q^2 = 0.728	ACC = 0.891; R^2 = 0.697	Drug discovery	(p46)

Abbreviations: ACC, accuracy; ANN, artificial neural network; XGBoost, extreme gradient boosting; GBM, gradient boosting machine; LR, logistic regression; BGR, bagging regressor.; ETR, extra trees regressor; GPR, Gaussian process regression; RP, recursive partitioning; RVM, relevance vector machine.

TABLE 2

Detailed information on representative toxicity prediction studies based on DL

Category	Toxicity endpoint	DL framework	Internal validity	External validity	Application area	Refs
Single task	Tox21 (7 NR, 5 SR)	GCNN	AUC = 0.687–0.830	AUC = 0.757	Drug discovery	[p27]
	Liver toxicity	DCNN	SE = 0.790; SP = 0.520	AUC = 0.670	Drug discovery	[p49]
	Hepatocellular hypertrophy	DNN	AUC = 0.830	AUC = 0.720	Drug discovery	[p50]
	Hematotoxicity	GCNN, DMPNN, attentive FP	AUC = 0.768	AUC = 0.759	Drug discovery	[p59]
	Ototoxicity	TEXTCNN, transformer CNN	AUC = 0.930/0.940	AUC = 0.900/0.870	Drug discovery	[p66]
	DILI	CNN	–	AUC = 0.960	Drug discovery	[p78]
	Seven types of human organ toxicity	DMPNN	AUC = 0.700–0.800	AUC = 0.640–0.920	Drug discovery	[p79]
	hERG	DNN, GCN	ACC = 0.812	ACC = 0.773	Drug discovery	[p80]
	hERG	DNN, LSTM	BACC = 0.800–0.840	BACC = 0.720–0.760	Drug discovery	[p81]
	Hematotoxicity	TEXTCNN algorithm, CNN fingerprint, transformer CNN	AUC = 0.630–0.730	AUC = 0.620–0.680	Drug discovery	[p58]
	hERG	DMPNN	AUC = 0.956	AUC = 0.810–0.849	Drug discovery	[p82]
	hERG	FCNN, GCNN, CNN	ACC = 0.860, AUC = 0.930	ACC = 0.746–0.810	Drug discovery	[p83]
	hERG	DNN	AUC = 0.872–0.889	ACC = 0.814–0.992	Drug discovery	[p84]
	DILI	GNN	–	ACC = 0.773	Drug discovery	[p85]
	DILI	GAN	ACC = 0.814	ACC = 0.787/0.765	Drug discovery	[p86]
	Mutagenicity	GCNN	ACC = 0.910, AUC = 0.950	ACC = 0.850, AUC = 0.880	Drug discovery	[p88]
	Mutagenicity	GCNN	AUC = 0.878	AUC = 0.838	Drug discovery	[p89]
	Carcinogenesis	GNN	–	MSE = 0.71; AUC = 0.730	Drug discovery	[p90]
	Acute toxicity	FFNN, GCNN, CNN	R^2 = 0.652–0.866	R^2 = 0.687–0.861	Drug discovery	[p92]
	Aquatic toxicity (three groups of organisms)	Transformer + DNN	R = 0.640–0.730	MAE = 1.97–6.04	Environmental safety	[p87]
	Chemical toxicity (27 endpoints)		AUC = 0.770–0.934	AUC = 0.886	Drug discovery	[p93]
	Skin corrosion/irritation	Attentive FP	AUC = 0.963	AUC = 0.891	Drug discovery	[p94]
Multitask	Tox21 (7 NR, 5 SR)	DNN	AUC = 0.700–0.930 AUC (NR.AR) = 0.350	–	Drug discovery	[p25]
	33 toxicity endpoints	Multitask graph attention framework	–	AUC = 0.618–0.981	Drug discovery	[p26]
	Ames mutagenicity	Multitask learning DNN	ACC = 0.730–0.900	ACC = 0.990	Drug discovery	[p32]
	Acute toxicity	Multitask DNN, Single-task DNN, RF, DL consensus architecture, GCNN	R^2 = 0.652–0.866	R^2 = 0.687–0.861	Drug discovery	[p92]
	Tox21 (7 NR, 5 SR)	Multitask DNN, multitask GCNN	AUC = 0.903, MCC = 0.580, Precision = 0.814	AUC = 0.897/0.912	Drug discovery	[p96]
	Tox21 (7 NR, 5 SR); clinical toxicity; RTECS *in vivo* toxicity	Multitask DNN	AUC = 0.991–0.994	AUC = 0.820–0.829	Drug discovery	[p97]
	hERG	Multitask DNN	AUC = 0.792–0.944	AUC = 0.883–0.967	Drug discovery	[p98]

Abbreviations: BACC, balanced accuracy; DCNN, deep convolutional neural network; FCNN, fully convolutional neural network; FFNN, feedforward neural network; MAE, mean absolute error; MLPR, multilayer perceptron regression; MSE, mean squared error; RTECS, Registry of Toxic Effects of Chemical Substances; SE, standard error; NR, nuclear receptor; SR, stress response.

plication area Refs
ug discovery [p27]
ug discovery [p49]
ug discovery [p50]
ug discovery [p59]
ug discovery [p66]
ug discovery [p78]
ug discovery [p79]
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ug discovery [p98]
perceptron regression; MSE, mean squared

已经应用了各种方法进行毒性预测，支持向量机（SVM）和随机森林（RF）由于其独特优势，已经成为当前基于数据驱动的毒性预测研究中最重要的两种传统机器学习（TML）方法，这一发现可以为后续相关研究提供有效参考。

DL

随着毒性数据的激增和计算能力的不断优化，深度学习（DL）作为机器学习（ML）的一个重要子集，凭借其能够自动构建复杂特征的独特优势，已成为毒性预测领域最有前景的技术之一。[p72],[p73]

深度学习（DL）也被称为深度神经网络（DNN），与人工神经网络（ANN）密切相关，后者是一种受人脑生物神经网络启发的机器学习算法（TML）。与只有三层（输入层、隐藏层和输出层）的ANN不同，DL算法具有多于一层的隐藏层，因此更适合分析具有复杂机制的大量毒理学数据。通过逐层提取和抽象特征，DL算法可以表征数据，并从大量训练样本中学习统计规律，从而能够基于未知数据进行更准确的预测。经典的DL算法主要包括卷积神经网络（CNNs）、循环神经网络（RNNs）、生成对抗网络和强化学习（RL），它们在人脸/语音/模式识别、自然语言处理、生物信息学以及药物发现等领域取得了重大进展。

近年来，许多基于数据的毒性预测研究已经证实了这样一种观点：当训练集足够大时，深度神经网络（DNN）可以优于人工神经网络（ANN）及上述所有传统机器学习（TML）方法。[p25],[p26],[p49] 我们将使用深度学习（DL）的研究分为两类：单任务预测和多任务预测，并在表2中列出了相关简要信息。从该表中可以看出，不同的深度学习算法被提出用于构建毒性预测模型，并取得了一些令人满意的结果。在毒性预测领域，流行的深度学习框架主要包括深度神经网络（DNN）、图注意力网络（GAN）、图神经网络（GNN）、图卷积网络（GCN）、长短期记忆网络（LSTM）、定向消息传递神经网络（DMPNN）以及注意力指纹（FP）。各种毒性端点预测研究都受益于这一系列深度学习方法，其中最具代表性的是多个被深入研究的毒性端点

GCN 是一种用于处理图数据的深度学习模型。传统的 CNN 适合处理规则的网格型数据，例如图像数据，但无法很好地处理不规则的图数据，例如社交网络和推荐系统中的用户–物品关系。通过将卷积操作扩展到图结构数据，GCN 能够有效地处理图数据中的节点和边。GCN 最突出的一点是，即使在未经训练且仅使用随机初始化参数的情况下，它提取的特征质量仍然足够高。LSTM 是一种设计用于捕捉序列数据中长期依赖关系的 RNN 架构。传统 RNN 在学习和保留长序列信息时可能会遇到梯度消失问题，但 LSTM 通过引入记忆单元，可以选择性地学习、存储和遗忘信息，从而专门解决这一问题。在实际应用中，LSTM 网络尤其...

除了hERG阻滞剂之外，Kang等人在2020年提出了一个用于药物性肝损伤（DILI）预测的模型，命名为GLIT，该模型使用转录响应数据和化学结构。GLIT使用生成对抗网络（GAN）学习嵌入向量，这是最有效的图神经网络（GNN）架构之一，通过节点及其邻居在图中的注意力分数来学习每个节点的隐藏表示。在该架构中，药物嵌入向量从一个三层神经网络的最后一层提取，药物诱导的基因表达嵌入向量则通过使用称为OmniPath的生物知识图谱的GNN提取。将这两个嵌入向量与药物给药剂量和持续时间信息拼接后，输入到GLIT的预测层以预测DILI。GLIT实现了0.788的F1值，优于基线模型。同样用于DILI预测，Huang等人使用多视图（MV）GNN和适当的数据增强策略来获得一个

KEYNOTE (GREEN)

various methods have been applied for toxicity prediction, SVM and RF have already become the two most important TML methods in data-driven toxicity prediction research in the current context based on their unique advantages, and this finding can serve an effective reference for subsequent related studies.

DL

With the proliferation of toxicity data and the continuous optimization of computing power, DL, as an important subset of ML, has become one of the most promising techniques in the field of toxicity prediction owing to its unique advantages that allow complex features to be constructed automatically.[p72],[p73] DL is also known as deep neural network (DNN) and is closely related to the ANN, a TML algorithm inspired by biological neural networks in the human brain. Unlike the ANN, which has only three layers (an input layer, a hidden layer, and an output layer), DL algorithms have more than one hidden layer and are therefore more suited for analyzing large amounts of toxicological data with complex mechanisms.[p22] Through the layer-wise extraction and abstraction of features, DL algorithms can characterize data and learn statistical laws from massive training samples and can thus make more accurate predictions based on unknown data. Classical DL algorithms mainly include convolutional neural networks (CNNs), recurrent neural networks (RNNs), generative adversarial networks, and reinforcement learning (RL), which have made great progress in the fields of face/voice/pattern recognition, natural language processing, bioinformatics, and drug discovery.[p78]

In recent years, many data-driven toxicity prediction studies have confirmed the notion that, when the training set is large enough, DNNs can perform better than ANNs and all the above-mentioned TML methods.[p25],[p26],[p49] We classified studies using DL into two categories, single-task and multitask predictions, and list brief information thereon in Table 2. From this table, we can see that different DL algorithms have been proposed to construct toxicity prediction models, and they have yielded some satisfactory results. Popular DL frameworks in the field of toxicity prediction mainly include the DNN, graph attention network (GAN), graph neural network (GNN), graph convolutional network (GCN), long short-term memory (LSTM) network, directed message passing neural network (DMPNN), and attentive fingerprint (FP). A variety of toxicity endpoint prediction studies have benefited from this raft of DL approaches, the most representative of which are several well-studied toxicity endpoints with relatively abundant data such as Tox21 pathway toxicity,[p79] hERG blocker toxicity,[p80],[p81] hematotoxicity,[p58],[p59] and so on. For single-task studies, DL methods have been applied to a specific toxicity endpoint, and most of the derived models have excellent predictive ability with evaluation metrics over 0.8. For example, as an important endpoint, the toxicity of hERG blockers has recently been predicted by most popular DL methods such as the DNN, GCN, LSTM, and DMPNN, obtaining satisfactory area under the receiver operating characteristic curve (AUC) values of up to 0.956 and 0.849 for internal and external validation, respectively.[p82],[p83],[p84] Among them, the GCN is a DL model for processing graph data. The traditional CNN is suitable for processing regular grid-type data, such as image data, but cannot deal well with irregular graph data, such as user–item relationships in social networks and recommendation systems. By extending the convolution operation to graph structure data, the GCN can effectively process nodes and edges in graph data. The most outstanding feature of the GCN is that the features it extracts are of sufficient quality even without training and using only randomly initialized parameters. LSTM is a type of RNN architecture designed to capture long-term dependencies in sequential data. Traditional RNNs can struggle to learn and retain information over long sequences due to the vanishing gradient problem, but LSTMs are specifically designed to address this issue by introducing a memory cell that can selectively learn, store, and forget information over time. For real-word application, LSTM networks are particularly effective in tasks that require the modeling of complex temporal patterns and long-range dependencies in data sequences. The MPNN is a class of supervised learning framework that can be applied to graphs. This framework abstracts some commonalities from the popular neural network models that support graph data, such as gate GNNs and graph attention networks. Unlike the traditional MPNN framework, where messages are passed from one node to its neighbors via the edges between them, the DMPNN is unique in that edges are directed, and edges, instead of atoms, act as the source and receiver of messages. These two features prevent information from being repeatedly passed back to its source, thus reducing noise. Recently, the DMPNN has been widely used for molecular property prediction because of its high efficiency and capacity for chemical representation.

In addition to the hERG blockers, in 2020, Kang *et al*. proposed a prediction model for DILI, named GLIT, using transcriptional response data and chemical structures.[p85] GLIT learns the embedding vectors using the GAN, one of the most effective GNN architectures, which learns hidden representations of each node using the attention scores of a given node and its neighbors in a graph. In this architecture, a drug embedding vector is extracted from the last layer of a neural network with three layers, and a drug-induced gene expression embedding vector is extracted from a GNN using a biological knowledge graph called OmniPath. The two embedding vectors are concatenated with the dosage and duration of drug administration information, and fed to the prediction layers of GLIT to predict DILI. GLIT achieved an F1 value of 0.788, which is better than those baseline models. Also for DILI prediction, Huang *et al*. used a multi-view (MV)-GNN and appropriate augmentation strategy to obtain a DILI model with a cross-validation accuracy of 0.814 in 2021. The MV-GNN model can extract atom messages and bond messages simultaneously. It considers atom message passing and bond message passing as two parallel streams and allows the atom/bond messages to communicate during the passing phase. It takes advantage of the two perspectives and thus outperforms other baseline models trained on DILI datasets in extensive experiments.[p78],[p85],[p86] Hematotoxicity, a serious but overlooked toxicity in discovery, has also attracted the attention of deep-learning researchers in recent years. In 2022,

KEYNOTE (GREEN)

Cao 等人构建了一个基于注意力 FP 的预测模型，该模型在验证集上获得了 72.6% 的平衡准确率和 76.8% 的 AUC 值。注意力 FP 是一种注意力机制，它为不同的输入特征分配不同的权重，以更多关注与目标相关的特征并抑制与目标无关的特征。这种机制常用于神经网络中，以提升模型的性能和泛化能力。具体而言，它可以应用于神经网络的每一层，通过动态重新分配权重来强化目标相关信息并抑制无关或冗余信息。注意力 FP 被提出作为一种强大的注意力机制，有助于提升神经网络的性能和泛化能力，尤其是在处理具有复杂关系和众多特征的毒性数据时。最近，Kristiansson 等人提出了一个使用变压器的水生毒性预测模型，以捕捉

除了单任务模型之外，多任务预测在深度学习背景下也得到了积极发展。多任务学习是一种机器学习方法，它将多个相关任务结合起来，基于共享表示进行学习。它涉及多相关任务的并行学习，并通过浅层共享表示相互帮助以提高泛化能力。从表2可以看出，由于其子数据集的信息共享机制，Tox21通路毒性是多任务学习的热门选择。2015年，Thomas Unterthiner的研究团队首次将多任务深度学习应用于Tox21数据挑战中的毒性预测，并赢得了两个小组挑战赛（核受体和应激反应），以及总体大奖赛，这标志着深度学习用于预测化学毒性的开端。鉴于不同子挑战/任务之间的高相关性，研究人员得出结论，多任务学习方法

分子输入行条目系统（SMILES）嵌入被用作多任务模型的输入。随后在其他一些毒性终点上的多任务研究也支持这一观点。2019年，Cheng 等人开发了一种用于药物诱导心脏毒性的有效预测工具，称为 deephERG，基于 7889 个化合物和多任务 DNN 算法。验证结果显示，deephERG 优于使用单任务 DNN、NB、SVM、RF 和图卷积神经网络（GCNN）构建的模型。具体而言，deephERG 最佳模型在验证集上的 AUC 值为 0.967。2021 年，Hou 等人提出了一种新颖的多任务图注意力（MGA）框架，可同时学习回归和分类任务，并使用 33 个毒性数据集确认了多任务预测的优越性。更近期，Martínez 等人使用 DL MTL 方法和来自五个菌株的实验信息建立了一种用于 Ames 测试的新模型，其结果超越了现有模型。

尽管深度学习（DL）模型在使用大数据预测毒性终点方面的优势令人惊讶，但对于数据量较小的终点，模型性能的提升非常有限。在过去几年中，DL模型也被用于构建一些研究较少的毒性终点的预测模型，但由于数据集相对较小，结果通常不太令人满意，或者仍然不如传统机器学习（TML）。[p50]，[p58],[p65]

2018 年，Tohkin 等人基于原始风险评估报告数据库（ORAD）、危险评估支持系统集成平台（HESS）以及一个合并数据集，使用深度学习（DL）、随机森林（RF）和支持向量机（SVM）开发了肝细胞肥大预测分类模型。研究发现，使用 HESS 数据集的 SVM 模型表现最佳，该模型使用 251 种化学物质进行训练，并预测了应用领域内的 214 种测试化学物质。类似地，2021 年，Li 等人整理了一个包含 2807 个分子的药物性耳毒性数据集，并使用五种传统机器学习（TML）和两种深度学习（DL）算法构建了一系列预测模型。内部和外部验证结果表明，基于 SVM 和 RF 的模型表现优于两种 DL 模型，最终基于六种 TML 模型获得了一致性模型。他们在同年进行的一项血液毒性预测研究中得出了类似的结论。在这些情况下，DL 算法与 SVM 和 RF 相比并未显示出显著优势。

Cao *et al*. constructed a prediction model based on attentive FP that yielded a balanced accuracy of 72.6% and an AUC value of 76.8% for the validation set.[p59] Attentive FP is an attention mechanism that assigns different weights to different input features to pay more attention to target-relevant features and suppress target-irrelevant features. This mechanism is commonly used in neural networks to improve the performance and generalization of a model. Specifically, it can be applied in each layer of the neural network to enforce target-related information and suppress irrelevant or redundant information by dynamically redistributing weights. Attentive FP is proposed as a powerful attention mechanism that helps to improve the performance and generalization ability of neural networks, especially when dealing with toxicity data with complex relationships and many features. Recently, Kristiansson *et al*. presented an aquatic toxicity prediction model using transformers to capture toxicity-specific features directly from chemical structures and obtained high predictive performance. Their results have demonstrated that DL and transformers constitute an adaptive data-driven approach for estimating structure–toxicity relationships that improves the predictive performance of existing QSAR methods, which may have some new implications for data-driven toxicity prediction.[p87] In addition to the toxicity properties mentioned above, these DL methods have also been successfully applied to some other toxicity endpoints, such as alternariol monomethyl ether (AME) toxicity,[p88],[p89] carcinogenesis,[p90],[p91] acute toxicity and so on.[p92],[p93],[p94]

As well as single-task models, multitask prediction has also been vigorously developed in the context of DL.[p95] Multitask learning is an ML method that combines multiple related tasks to learn based on a shared representation. It involves the parallel learning of multiple related tasks at the same time, and multiple tasks help each other learn through shallow shared representations to improve generalization. From Table 2, we can see that Tox21 pathway toxicity is a popular choice for multitask learning because of the information-sharing mechanism of its sub-datasets. In 2015, the research group of Thomas Unterthiner first applied multitask DL to toxicity prediction in a Tox21 data challenge and won both panel challenges (nuclear receptors and stress response) as well as the overall Grand Challenge, which 'kicked off' the use of DL to predict chemical toxicity. Given the high correlations between different subchallenges/tasks, the researchers concluded that the multitask learning approach significantly outperforms the single-task network in almost all tasks.[p25] Recently, Liu *et al*. developed three multitask models for 12 Tox21 datasets based on a descriptor, FP, and molecular graph, respectively. Then, they proposed a comodel based on averaging for balance, and to make more accurate predictions, finally obtaining a comodel with an AUC of 0.903 and 0.897 for cross- and external validation, respectively.[p96] Moreover, Das *et al*. have also demonstrated the advantage of employing a multitask model to predict toxicity across *in vitro*, *in vivo*, and clinical platforms and concluded that there is a minimal relative importance for *in vivo* data (animal data) for predicting clinical toxicity, in particular when unsupervised pretrained simplified molecular input line entry system (SMILES) embeddings are used as input to multitask models.[p97] Subsequent multitask research on some other toxicity endpoints has also supported this opinion. In 2019, Cheng *et al*. developed an effective prediction tool for drug-induced cardiotoxicity, termed deephERG, based on 7889 compounds and a multitask DNN algorithm. Validation results showed that deephERG outperformed models built using a single-task DNN, NB, SVM, RF, and graph convolutional neural network (GCNN). Specifically, the AUC value for the best deephERG model was 0.967 on the validation set.[p98] In 2021, Hou *et al*. proposed a novel multitask graph attention (MGA) framework to learn regression and classification tasks simultaneously and, using 33 toxicity datasets, confirmed the superiority of multitask prediction.[p26] More recently, Martínez *et al*. built a novel model for the Ames test using a DL MTL approach and experimental information from five strains, and the results surpassed those obtained by single-task models.[p32] Above all, these multitask studies have revealed the following advantages of the multitask model over the single-task model: (1) compared with single-task models, multitask models are more efficient, especially in the context of big data. (2) the multitask approach allows the capture of possible correlations between toxicity endpoints and the sharing of hidden unit representations among prediction tasks, which is particularly important for some tasks where very few measurements are available. Moreover, a further study based on 200 biological targets also found that adding additional tasks and data can improve the predictive power of a multitask DNN model.[p99]

Although the advantage of DL models for predicting toxicity endpoints with big data is surprising, the improvement in model performance is very limited for those endpoints with small datasets. In the last few years, DL models have also been used to construct prediction models for some less-studied toxicity endpoints, but the results were usually less satisfactory or still inferior to TML due to their relatively small datasets.[p50],[p58],[p65] In 2018, Tohkin *et al*. developed predictive classification models for hepatocellular hypertrophy using DL, RF, and SVM based on the Original Risk Assessment reports Database (ORAD), the Hazard Evaluation Support System Integrated Platform (HESS), and a combined dataset. The best model was found to be the SVM model using the HESS dataset, which was trained with 251 chemicals and predicted 214 test chemicals inside the application domain.[p50] Similarly, in 2021, Li *et al* collated a drug-induced ototoxicity dataset with 2807 molecules and constructed a series of prediction models using five TML and two DL algorithms. The results of internal and external validation showed that models based on SVM and RF performed better than two DL models, and a consensus model was finally obtained based on six TML models.[p65] They drew a similar conclusion in a hematotoxicity prediction study conducted in the same year.[p58] In these cases, DL algorithms did not show significant advantages compared with some outstanding TML methods, such as RF and SVM. The reason for this is that the main advantage of DL is embodied in big data, which can be used to automatically obtain high-level features using a large amount of

训练数据。因此，我们假设当有更多训练数据可用时，大多数深度学习（DL）算法将优于传统的机器学习（ML）方法。总之，尽管深度学习在毒性预测研究中的应用已经得到了极大推进并取得了显著的成果，但在使用深度学习模型进行毒性预测时，主要的困难仍然是训练数据不足。

PDL

鉴于数据驱动毒性预测领域中数据稀疏带来的困难，以及在预测任务中不总是可能或容易增加更多标注数据或构建多任务模型的事实，近年来，研究人员提出了一些应对“少数据”困境的可行方法。为了减少深度学习（DL）模型对大量毒性数据的依赖，Siu 等人在 2021 年将半监督学习（SSL）方法应用于 Tox21。SSL 是一种混合学习方法，适用于只有部分数据带有标注的数据学习任务，而且标注数据的数量远小于未标注数据的数量。与监督学习相比，它是一种可以通过少量标注数据和额外的未标注样本作为辅助来提高学习性能的范式，并提供了一种通过额外未标注样本探索潜在模式的方法，从而减少对大量标注数据的需求。

SSL 已在包括生物信息学、图像分类、语音识别等多个领域得到广泛应用。在本研究中，作者假设标签函数在高密度区域内是平滑的，因此位于同一区域的数据点应共享相同的标签。他们提出了一种用于预测化学毒性的 GCN，并通过平均教师（MT）SSL 算法训练该网络。结果显示，SSL 模型的平均 AUC 为 0.757，比通过监督学习和传统机器学习方法训练的模型提高了 6%。受到这一成功的鼓舞，预计 SSL 通过利用现有的大型无标签化学数据库，将有助于其他化学性质预测任务。

迁移学习（TL）是一种机器学习方法，它在数据丰富的环境中学习执行某项任务，然后将所学知识“迁移”到数据较少的任务中。这种方法在解决关联问题有丰富标注数据但研究问题数据不足的情况下具有重要价值。目前，TL 被用于解决许多领域的低数据任务，例如计算机视觉、自然语言处理和药物发现。当前的毒性预测研究涉及许多复杂的机制和途径，因此 TL 在深度学习环境中具有极大的潜力来提高模型性能。例如，Tung 等人开发了一种基于集成树的多任务学习方法，将皮肤致敏的明确定义的 AOP 中的三个相关任务结合起来，以将共享知识迁移到理解人类致敏剂的主要任务中。他们的模型与传统的单一任务模型相比显示出更优的性能。

敏感性。[p108] 一个整合的贝叶斯模型在“人类数据集”上进行了训练，使用了为五种皮肤敏感性检测开发的模型的预测结果，这些结果被描述为描述符，并结合NB算法，模型达到了0.89的预测准确率。实际上，迁移学习（TL）从根本上不同于半监督学习（SSL）和多任务学习方法。具体来说，TL的源数据集和目标数据集可以具有不同的分布，只要它们彼此相关，而在SSL中，源数据和目标数据来自同一个数据集，并且目标数据集没有标签。

多任务学习旨在同时解决多个任务，并学习一个对所有研究任务都表现良好的共享表示/模型。相比之下，在迁移学习（TL）中，目标领域是重点，知识已经从源数据中获得以用于目标数据，且这些领域不需要相关或同时发挥作用。[p109] 一些示例还表明，对于迁移学习而言，领域相关性比数据量更为重要。[p110] 近年来，与迁移学习相关的研究发展迅速；尽管迁移学习的应用仍处于初期阶段，还有许多问题有待研究，但在小数据的背景下，它可能很快会变得更加高效和广泛使用。此外，人工数据生成和强化学习（RL）也可能是从小数据中挖掘有用信息的有前景的工具。

除了SSL和TL之外，属性增强是小数据建模的另一种有效方法。为了利用深度图学习模型的表达能力，Huang等人在2021年提出了一种属性增强策略，通过包括大量训练数据和其他信息，获得适用于DL算法的合格DILI数据集。基于初始DILI数据集与hERG、Ames、线粒体膜电位及磷脂沉积数据集之间存在分子重叠的事实，作者试图利用这些额外的毒性信息来提升DILI的预测性能。具体而言，他们基于药物的SMILES表示将这五个数据集合并，然后采用多标签训练方法建立属性增强学习过程。大量实验表明，该方法在DILI数据集上显著优于所有现有基线。最近，Wang等人也提出了类似研究。

总之，基于深度学习（DL）的最新毒性预测研究强调了这样一个事实：深度学习能够，并且实际上已经在，通过其强大的学习能力对毒性预测领域产生重大影响。它不仅可以生成高质量的单任务和多任务预测模型，还可以发现新的或潜在的结构警示，这对旨在从毒性数据集中挖掘毒性信息的药物化学家大有帮助。为了解决深度学习环境下的小数据相关问题，一些新方法，包括自监督学习（SSL）、迁移学习（TL）和性质增强学习已经被提出并取得了令人满意的成果。尽管深度学习在应用中仍存在一些困难，例如

training data. Therefore we hypothesize that most DL algorithms will outperform traditional ML methods when more training data are available. In summary, although the application of DL in toxicity prediction research has been greatly advanced and achieved remarkable results, the main difficulty when using DL models for toxicity prediction is still insufficient training data.

The PDL

Given the difficulty posed by data sparsity in the field of data-driven toxicity prediction, and the fact that it is not always possible or easy to add more annotated data to a prediction task or construct multitask models, researchers in recent years have proposed some viable approaches to the 'small data' dilemma. To reduce the dependency of DL models on extensive toxic data, the semisupervised learning (SSL) method was applied to Tox21 by Siu *et al*. in 2021. SSL is a hybrid approach for learning tasks where only a portion of the data are labeled, and the amount of labeled data is much smaller than that of unlabeled data. It is a paradigm that can improve learning performance using a few labeled data and additional unlabeled examples as auxiliaries compared with supervised learning, and it provides a way to explore latent patterns through additional unlabeled examples, alleviating the need for a large number of labels.[p100] SSL has found wide applications in various fields including bioinformatics, image classification, speech recognition, and many others.[p101] In this study, the authors assumed that the label function is smooth in high-density areas, so data points located in the same area should share the same label. They proposed a GCN to predict chemical toxicity and trained the network via the mean teacher (MT) SSL algorithm. The results showed that the average AUC of the SSL model was 0.757, which is a 6% improvement over the model trained by supervised learning and conventional ML methods.[p27] Encouraged by this success, SSL is expected to be beneficial to other chemical property prediction tasks through utilizing existing large unlabeled chemical databases.

Transfer learning (TL) is an ML approach that learns to perform a task in a data-rich environment and then 'transfers' the learned knowledge to a task with fewer available data. This approach is of great value for resolving situations where there is abundant labeled data for an associated problem but insufficient data for the problem under study.[p102] Presently, TL is used to address low-data tasks in many fields such as computer vision,[p103] natural language processing,[p104] and drug discovery.[p105],[p106] Current toxicity prediction studies involve many complex mechanisms and pathways, and TL thus has great potential to improve model performance in a DL environment. For example, Tung *et al*. developed an ensemble tree-based multitask learning method incorporating three relevant tasks in the well-defined AOP of skin sensitization to transfer shared knowledge to the major task of understanding human sensitizers. Their model showed superior performance compared with conventional single-task methods and the other four existing skin sensitization prediction methods.[p107] Similarly, in 2020, Andrade *et al*. updated PredSkin to integrate multiple models developed with *in vitro*, *in chemico*, and mouse and human *in vivo* data, integrated into a consensus NB model that predicts human skin sensitivity.[p108] An integrative Bayesian model was trained on the 'Human dataset' using the predictions of models developed for five skin sensitization assays, described as descriptors, along with the NB algorithm and achieved a prediction accuracy of 0.89. In fact, TL is fundamentally different from SSL and multitask learning methods. Specifically, the source and target datasets of TL can have a different distribution and merely be related to each other, whereas in SSL, the source and target data are from the same dataset, and the target set does not have the labels.

Multitask learning aims to solve multiple tasks simultaneously and to learn a shared representation/model that performs well for all studied tasks. In contrast, in TL, the target domain is the focus, knowledge has already been obtained for target data from source data, and the domains do not need to be related or to function simultaneously.[p109] Several examples have also shown that domain relatedness is more important than data size for TL.[p110] In recent years, TL-related research has developed rapidly; although the application of TL is still in its infancy, and there is still much to be studied, it may soon be more effective and widely used in the context of small data. Furthermore, artificial data generation and RL may also be promising tools to mine useful information from small data.

In addition to SSL and TL, property augmentation is another effective approach to small data modeling. To take advantage of the expressive power of deep graph learning models, in 2021, Huang *et al*. proposed a property augmentation strategy to include massive training data along with other information to obtain a qualified DILI dataset for a DL algorithm. Based on the fact that there are overlapping molecules between initial DILI and hERG, Ames, mitochondrial membrane potential, and phospholipidosis datasets, the authors sought to utilize this additional toxicity information to enhance the prediction performance of DILI. Specifically, they combined these five datasets based on a SMILES representation of the drugs and then employed a multilabel training approach to establish a property augmentation learning process. Extensive experiments demonstrated that the proposed method significantly outperformed all existing baselines on the DILI datasets. More recently, Wang *et al*. also proposed a framework that combines feature selection, TL, and data augmentation for toxicity prediction and classification based on a small sample size. They obtained augmented data by adding white noise and then utilized these data together with the source domain data to train a CNN.[p111] Property augmentation learning can provide more input data to better learn molecular vector representations and thus is a promising approach to improve the prediction performance for toxicity endpoints with limited data available.[p86]

To sum up, recent toxicity prediction studies based on DL highlight the fact that DL can, and in fact already has, greatly impact the field of toxicity prediction with its powerful learning ability. It can not only generate high-quality single- and multitask prediction models but also discover new or potential structural alarms, which is a great help for medicinal chemists aiming to mine toxicity information from toxicity datasets. To solve the small data-related issues in the DL context, some novel methods including SSL, TL, and property augmentation learning have been proposed and achieved satisfactory results. Although there are still some difficulties in the application of DL, such as

TABLE 3

Detailed information on popular toxicity evaluation tools

	Endpoint	Methods	Accuracy	Toxic substructure	Explanation	URL
ADMETlab3.0	36 endpoints and 8 rules	DMPNN	0.67 ~ 0.97	Yes	+++	https://admetlab3.scbdd.com/
admetSAR3.0	55	CLMGraph	~0.87	No	+++	https://lmmd.ecust.edu.cn/admetsar3/
Deep-PK	21	DMPNN	0.16–0.89	Yes	+++	https://biosig.lab.uq.edu.au/deeppk/
HelixADMET	16	GNN	0.71 ~ 0.95	No	+	https://paddlehelix.baidu.com/app/drug/admet/train
pkCSM	10	GBS	0.68 ~ 0.90	No	++	https://biosig.unimelb.edu.au/pkcsm/prediction
vNN-ADMET	6	VNN	0.71 ~ 0.89	No	+	https://vnnadmet.bhsai.org/vnnadmet/login.xhtml
FP-ADMET	29	RF	0.64 ~ 0.89	No	+	https://gitlab.com/vishsoft/fpadmet
ICDrug	6	RF	–	No	+	https://www.icdrug.com/ICDrug/T
ProTox3.0	61	RF, DNN	0.75 ~ 0.93	No	+++	https://tox.charite.de
VenomPred2.0	8	RF, SVM, kNN, MLP	0.76 ~ 0.98	No	++	https://www.mmvsl.it/wp/venompred2/
AquaticTox	5	RF, AdaBoost, gradient boosting, SVM, FCNet, GACNN	0.71–0.8	No	+	https://chemyang.ccnu.edu.cn/ccb/server/ AquaticTox
eMolTox	14	RF	0.83 ~ 0.94	Yes	++	https://xundrug.cn/moltox
TEST	9	Consensus	–	No	+	https://www.epa.gov/chemical-research/toxicity-estimation-software-tool-test
Toxtree	15 rules	DT	–	Yes	++	https://apps.ideaconsult.net/data/ui/toxtree
VirtualRat	12	RF, DT	–	No	+	https://virtualrat.cmdm.tw/
Pred-Skin 3.0	1	NB	0.89	No	+	https://predskin.labmol.com.br/
Pred-hERG 5.0	1	SVM	0.8	No	+	https://predherg.labmol.com.br/
CarcinoPred-EL	1	XGBoost	0.7	No	+	https://ccsipb.lnu.edu.cn/toxicity/CarcinoPred-EL/
CardPred	1	ANN	0.9	No	+	https://bioanalysis.cau.ac.kr:7050/
CardioDPi	3	DNN	0.89–0.94	No	+	https://cardiodpi.sapredictor.cn/

Abbreviations: CLMGraph, contrastive learning based multitask graph neural network; FCNet, fusion-correction network, GBS, graph-based signatures; VNN, variable nearest neighbor, GACNN, convolutional neural networks based on genetic algorithm.

bdd.com/
du.cn/admetsar3/
edu.au/deeppk/
aidu.com/app/drug/admet/train
b.edu.au/pkcsm/prediction
sai.org/vnnadmet/login.xhtml
shsoft/fpadmet
com/ICDrug/T
t/wp/venompred2/
nu.edu.cn/ccb/server/
noltox
/chemical-research/toxicity-
tool-test
sult.net/data/ui/toxtree
lm.tw/
nol.com.br/
mol.com.br/
u.cn/toxicity/CarcinoPred-EL/
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eural networks based on genetic algorithm.

我们认为，深度学习（DL）将在标准毒性数据库的发展和计算机技术的更新下，为毒性预测带来新的机遇，尽管这仍然存在对足够平衡且大规模数据集的需求、缺乏透明性和解释性、程序需要长时间运行以及对高性能计算机的需求等问题。

预测软件和工具

受到各种毒性终点优秀预测模型不断积累以及毒理学家对实用研究结果迫切需求的鼓舞，一些用于单一或多个终点的预测软件和工具已经被开发出来。在本部分，我们整理了可免费访问的网络工具，并在表3中列出了它们的详细信息。

表3中详细列出了19种可用于毒性预测的可访问工具，并对其信息进行了仔细总结，包括建模方法、预测准确性以及预测结果的可解释性。所有网络服务器工具可以根据涉及的终点分为两类：综合吸收、分布、代谢、排泄和毒性（ADMET）系统[p112],(p113),(p114),(p115)。

(p116)、(p117)、(p118)、(p119) 以及个体毒性预测工具。[p108],(p120),(p121),(p122),(p123),(p124),(p125),(p126),(p127),(p128),(p129),(p130)

与毒性终点预测和建模方法的演变趋势类似，随着数据量的增加和方法的改进，预测工具也变得更加实用。在ADMET系统中，顶尖的有ADMETlab 3.0、admetSAR 3.0、Deep-PK和HelixADMET，它们在基于机器学习（ML）或深度学习（DL）的预测准确性方面相当（约0.9），但在预测结果的解释能力上有所不同。通常，在ADMETlab 3.0项目中，多任务DMPNN架构结合分子描述符不仅保证了每个终点同时分析的计算速度，还在准确性和稳健性方面表现出优越性能。除了基本的预测功能外，ADMETlab 3.0引入了应用程序接口（API），以满足对大量数据进行程序化访问日益增长的需求。为了进一步提高毒性预测工具的实用性，一个新模块也被嵌入。

他们在毒性研究中增加了一些更复杂的终点（例如肝毒性、肾毒性、TOX-21通路毒性）。

存储库并应用更可靠的机器学习方法（例如 RF、SVM 和 kNN）以获得更好的预测能力（准确率 =0.75 ～ 0.94）。此外，一些网络服务器工具专注于单一但极其重要的毒性终点，例如 CardPred[p129] 用于预测致癌性，Pred-hERG 5.0[p127] 用于预测心脏毒性，以及 PredSkin 3.0[p108] 用于预测皮肤敏感性。

总之，大多数用于毒性预测的最佳网络服务器工具是ADMET系统，随着大型数据库和计算机程序的加入，这些系统得到了改进和发展。尽管现有的用于单个毒性预测的软件包已经具有相当的预测能力和实用性，但通过应用新的深度学习方法构建更准确的模型或构建更预测更多毒性终点的模型，仍可进一步改进。我们相信，随着新算法的不断发展、标准毒性数据集的公开以及分子表示方法的进步，我们在不久的将来将能够获得更多科学且实用的个体毒性评估工具。

结束语

毒性评估是新药安全评价的重要组成部分，与人类健康和药物候选物的命运密切相关。近年来，数据驱动的毒性预测已经从传统机器学习（TML）发展到深度学习（DL），然后到多任务学习（PDL）。在此过程中，出现了一系列显著的预测平台，尤其是基于深度学习和多终端毒性预测的平台，这些平台具有更高的预测准确性和更广的适用性。本文回顾并讨论了毒性预测的特点与难点、建模方法的演变以及现有的预测工具。考虑到数据驱动毒性预测的研究现状，未来毒性预测可能有几个方向：（1）创建具有充分且可靠实验数据的基准毒性数据库；（2）构建更精细或机制明确的模型，例如基于已知AOP的QSAR模型；（3）开发新的m

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the need for a sufficiently balanced and large dataset, the lack of transparency and interpretation, the long period over which the program must be run, and the need for a high-performance computer, etc., we believe that DL will bring new opportunities for toxicity prediction with the development of standard toxicity databases and updates in computer technology.

Prediction software and tools

Encouraged by the continuous accumulation of excellent prediction models for various toxicity endpoints and the urgent demand from toxicologists for practical research results, some prediction software and tools have been developed for single or multiple endpoints. In this part, we collated the freely available web tools and list their detailed information in Table 3.

There are 19 accessible tools for toxicity prediction detailed in Table 3, and their information is carefully summarized including modeling methods, prediction accuracy, and the interpretability of their prediction results. All the webserver tools can be classified into two categories according to the endpoints involved: comprehensive absorption, distribution, metabolism, excretion, and toxicity (ADMET) systems[p112,p113,p114,p115,p116,p117,p118,p119] and individual toxicity prediction tools.[p108,p120,p121,p122,p123,p124,p125,p126,p127,p128,p129,p130] Similar to the evolutionary trends in toxicity endpoint prediction and modeling approaches, prediction tools have also become more practical as data have increased and methods have improved. Among the ADMET systems, the top ones are ADMETlab 3.0,[p112] admetSAR 3.0,[p113] Deep-PK,[p114] and HelixADMET,[p115] which have comparable prediction accuracy based on ML or DL (0.8 ~ 0.9) but differ in terms of the explanatory power of the prediction results. Typically, in ADMETlab 3.0 projects, the multitask DMPNN architecture coupled with molecular descriptors not only guarantees calculation speed for each endpoint simultaneously analyzed but also achieves superior performance in terms of accuracy and robustness. In addition to the basic prediction function, an application programming interface (API) has been introduced to meet the growing demand for programmatic access to large amounts of data in ADMETlab 3.0. To further improve the utility of the toxicity prediction tool, a new module was embedded in the web server of admetSAR 3.0, called ADMETopt, which can automatically replace other scaffolds in its scaffold library; this is very attractive to medicinal chemists who need to perform lead optimization. Deep-PK is a recently reported deep learn-based ADMET prediction platform that supports property prediction, molecular optimization, and interpretation to help users better understand the pharmacokinetics and toxicity of a given input molecule.[p114] Recently, Wang *et al.* developed a highly robust and extensible ADMET system called HelixADMET, which incorporates the concept of self-supervised learning and a GNN. It can not only make accurate predictions even for molecules with unobserved scaffolds but also provides users with a model fine-tuning interface, allowing them to build new and customized endpoints. As for individual toxicity prediction tools, those with multiple endpoints are more practical, such as ProTox3.0,[p122] eMolTox,[p120] and AquaticTox.[p124] They add some more complex endpoints (e.g. hepatotoxicity, nephrotoxicity, TOX-21 pathway toxicity) to the toxicity repository and apply more reputable ML approaches (e.g. RF, SVM, and kNN) to obtain better predictive power (accuracy = 0.75 ~ 0.94). Moreover, some webserver tools focus on single but exceptionally important toxicity endpoints, such as CardPred[p129] to predict carcinogenicity, Pred-hERG 5.0[p127] to predict cardiotoxicity, and PredSkin 3.0[p108] to predict skin sensitivity.

In summary, most of the best webserver tools for toxicity prediction are ADMET systems, which have improved and evolved with the addition of large databases and computer programs. Although existing software packages for making individual toxicity predictions already have considerable predictive power and practicability, they could be further improved by applying novel DL approaches to build more accurate models or building models predicting more toxicity endpoints. We believe that with the continuous development of new algorithms, the disclosure of standard toxicity datasets, and the progress of molecular representation methods, we will have access to more scientific and practical individual toxicity assessment tools in the near future.

Concluding remarks

Toxicity assessment is an important part of safety evaluation for new drugs, which is closely related to human health and the fate of drug candidates. In recent years, data-driven toxicity prediction has progressed from TML to DL, and then to the PDL. In this process, a series of remarkable prediction platforms have been generated, especially DL-based and multiple-endpoint toxicity prediction platforms with higher prediction accuracy and wider applicability. In this paper, we reviewed and discussed the features and difficulties of toxicity prediction, the evolution of modeling approaches, and the available predictive tools. Considering the research status of data-driven toxicity prediction, there are probably several directions for future toxicity prediction: (1) creating benchmark toxicity databases with sufficient and reliable experimental data; (2) constructing more elaborate or mechanistically explicit models, such as QSAR models based on known AOPs; (3) developing novel methods (e.g. new TL methods) to capture underlying information in small datasets more efficiently; (4) taking account of more interpretive or informative molecular attributes, such as descriptors that carry information about toxic targets; and (5) developing comprehensive toxicity prediction systems for various endpoints using advanced DLs. In conclusion, studies of toxicity prediction are updated annually, accompanied by improvements in data and computer programs. Although there are still some difficulties, we believe that data-driven toxicity prediction will soon move to a new stage and become an indispensable practical tool for drug safety assessment.

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利益声明

作者声明与本研究无任何利益冲突。

数据可用性

本文描述的研究未使用任何数据。

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CRediT authorship contribution statement

Ningning Wang: Writing – original draft. **Xinliang Li:** Data curation. **Jing Xiao:** Data curation. **Shao Liu:** Conceptualization. **Dongsheng Cao:** Conceptualization.

Declarations of interest

The authors declare no conflict of interest related to this study.

Data availability

No data was used for the research described in the article.

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